

AMENDMENT NO. 1 APRIL 1975
TO
IS : 6306 - 1971 METHODS OF TEST FOR
RECLAIMED RUBBER

Addendum

[*Page 13, RR : 5*) — Add the following alternative method for the determination of carbon black after 6.1 :

DETERMINATION OF CARBON BLACK
(ALTERNATIVE METHOD)

1. SCOPE

1.1 This method is intended for determination of carbon black in reclaimed rubber.

2. OUTLINE OF THE METHOD

2.1 The reclaimed rubber is heated in a furnace at about 1 000°C in an atmosphere of nitrogen to distill all other materials except ash and free carbon. Ash is also determined separately and from the difference of the two, amount of carbon black is obtained.

3. APPARATUS

3.1 Electrical Combustion Furnace — tubular in shape, about 325 mm in length and having a heating zone of about 275 mm, and capable of maintaining a temperature of $1\,000 \pm 25^\circ\text{C}$.

3.2 Combustion Tube — of transparent silica, about 725 mm in length, 26 and 28 mm in inner and outer diameter respectively.

3.3 Nitrogen Gas Cylinder

3.4 Gas Washing Bottles

3.5 Glazed Porcelain Boat

4. PROCEDURE

4.1 Ignite the boat and weigh it after cooling. Accurately weigh, in the boat, about 1 g of the homogenized sample prepared as in 3.1.1 (*see page 4*). Heat the furnace with the combustion tube to approximately 1 000°C. Draw out the combustion tube a little bit from the hot zone and clean it with nitrogen gas for 1 minute to drive away the air. Introduce the weighed boat with sample into the combustion tube, connect the gas flow and push the tube so that the boat is in the hot zone of the furnace. At first dense,

black-brown smoke will come out of the combustion tube which will slowly turn to white smoke. When white smoke becomes thinner, draw out the combustion tube from the furnace. Continue the flow of nitrogen for 2-3 minutes more. Cut off the supply of nitrogen and take out the boat carefully avoiding spilling and keep it in a desiccator till it is cooled. Weigh the boat with the ignited sample.

NOTE — Nitrogen gas used shall be cleaned by passing it through washing bottles containing glycerine and alkaline pyrogallol (20 – 30 g of pyrogallol in 10 ml of about 30 percent potassium hydroxide) respectively to remove the moisture and traces of oxygen present in nitrogen.

5. CALCULATION

5.1 Calculate the carbon black content as follows:

5.1.1 Total fillers,
percent by mass (A) = $\frac{M_3 - M_1}{M_2 - M_1} \times 100$

where

M_3 = mass in g of the boat with the sample after ignition,

M_1 = mass in g of empty boat, and

M_2 = mass in g of the boat with sample.

5.1.2 Ash content, percent by mass (calculated as given in RR : 3) = B.

5.1.3 Carbon black, percent by mass = A – B.

(CDC 42)

Printed at Dee Kay Printers, New Delhi